



449/1
DRAWING AND DESIGN
Paper 1
Nov. 2016
2½ hours



THE KENYA NATIONAL EXAMINATIONS COUNCIL
Kenya Certificate of Secondary Education
DRAWING AND DESIGN
Paper 1
2½ hours

Instructions to candidates

- (a) *You should have the following for this examination:*
 - Answer sheet;*
 - Drawing instruments;*
 - 3 sheets of drawing paper size A3;*
 - Scale rule.*
- (b) *This paper consists of **three** sections: **A**, **B** and **C**.*
- (c) *Answer **all** the questions in sections **A** and **B** and any other **two** questions from section **C**.*
- (d) *Questions in section **A** must be answered on the answer sheet provided.*
- (e) *Questions in section **B** and **C** should be answered on A3 sheets of drawing paper provided.*
- (f) *All dimensions are in millimetres unless otherwise stated.*
- (g) ***Candidates may be penalised for not following the instructions given in this paper.***
- (h) ***This paper consists of 10 printed pages.***
- (i) ***Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.***
- (j) ***Candidates should answer the questions in English.***



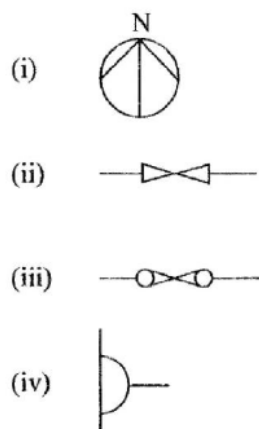


SECTION A (50 marks)

Answer **all** the questions in this section on the answer sheet provided.

1. (a) State **one** disadvantage of using each of the following items to hold paper on the drawing board.
- (i) Marking tape
 - (ii) Thumb pins (2 marks)
- (b) (i) Give **one** disadvantage of oral communication when representing an engineering object.
- (ii) Give **one** reason for observing established standards in manufacturing industry. (2 marks)

2. (a) Sketch each of the following lines:
- (i) Hidden detail
 - (ii) Folding line
 - (iii) Centre line
 - (iv) Cutting plane (2 marks)
- (b) State the meaning of each of the symbols shown in **Figure 1**.



(4 marks)

Figure 1



3. (a) **Figure 2** shows an elevation of a template.

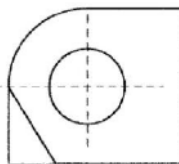


Figure 2

Measure the dimensions for the:

- (i) Circle
 - (ii) Radius
 - (iii) Angle of the slanting face (3 marks)
- (b) List six computer programs that can be used to produce a drawing. (3 marks)
4. (a) Define each of the following properties of materials:
- (i) plasticity
 - (ii) elasticity (2 marks)
- (b) State **four** ways in which design ideas are communicated. (2 marks)
5. (a) List **four** factors to consider when lettering. (2 marks)
- (b) State **three** effects of poor disposal of engineering materials to the environment. (3 marks)



916104

Kenya Certificate of Secondary Education, 2016
449/1

Turn over



6. Enlarge **Figure 3** (ABCD) in the ratio of 5:7

(4 marks)

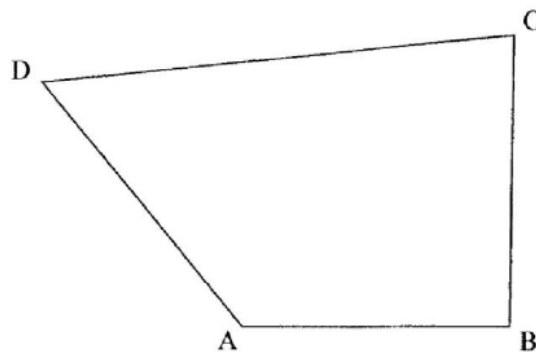


Figure 3

7. Construct a triangle whose perimeter is 240 mm and the sides are in the ratio 4.5:6.0:7.5. Measure the smallest angle.

(5 marks)

8. **Figure 4** shows the front elevation and incomplete plan of a truncated square-based pyramid.

- (a) complete the plan.
(b) draw the true shape of the cut face.

(5 marks)

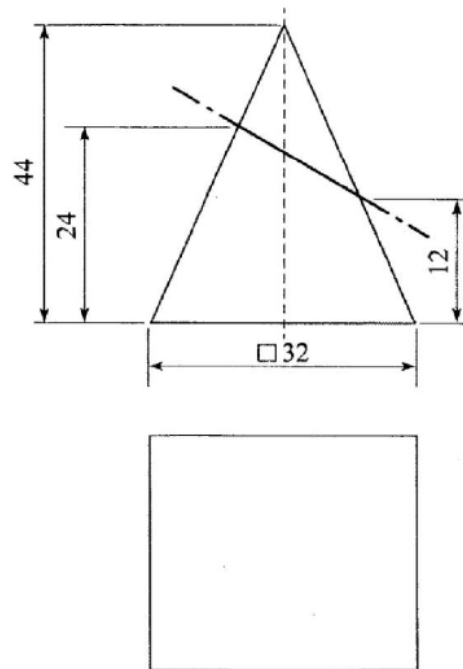


Figure 4



9. **Figure 5** shows two views of a machined bracket drawn in first angle project.

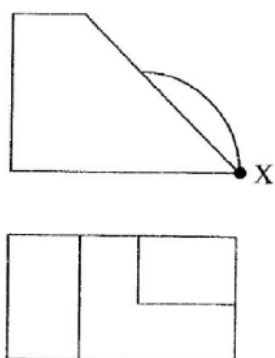


Figure 5

Sketch in good proportion, the isometric view of the block taking X as the lower point.

(5 marks)

10. **Figure 6** shows two views of a block drawn in first angle project. In good proportion sketch the block in oblique projection. (6 marks)

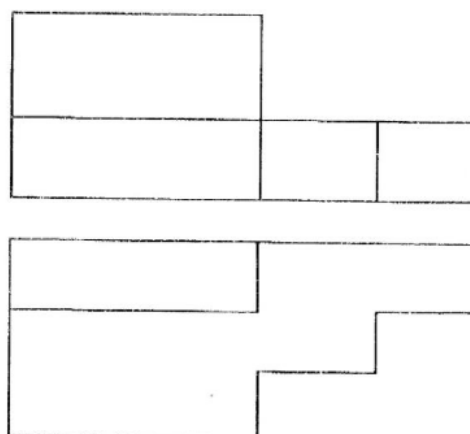


Figure 6



916104

Kenya Certificate of Secondary Education, 2016
449/1

Turn over



SECTION B (20 marks)

Question 11 is compulsory

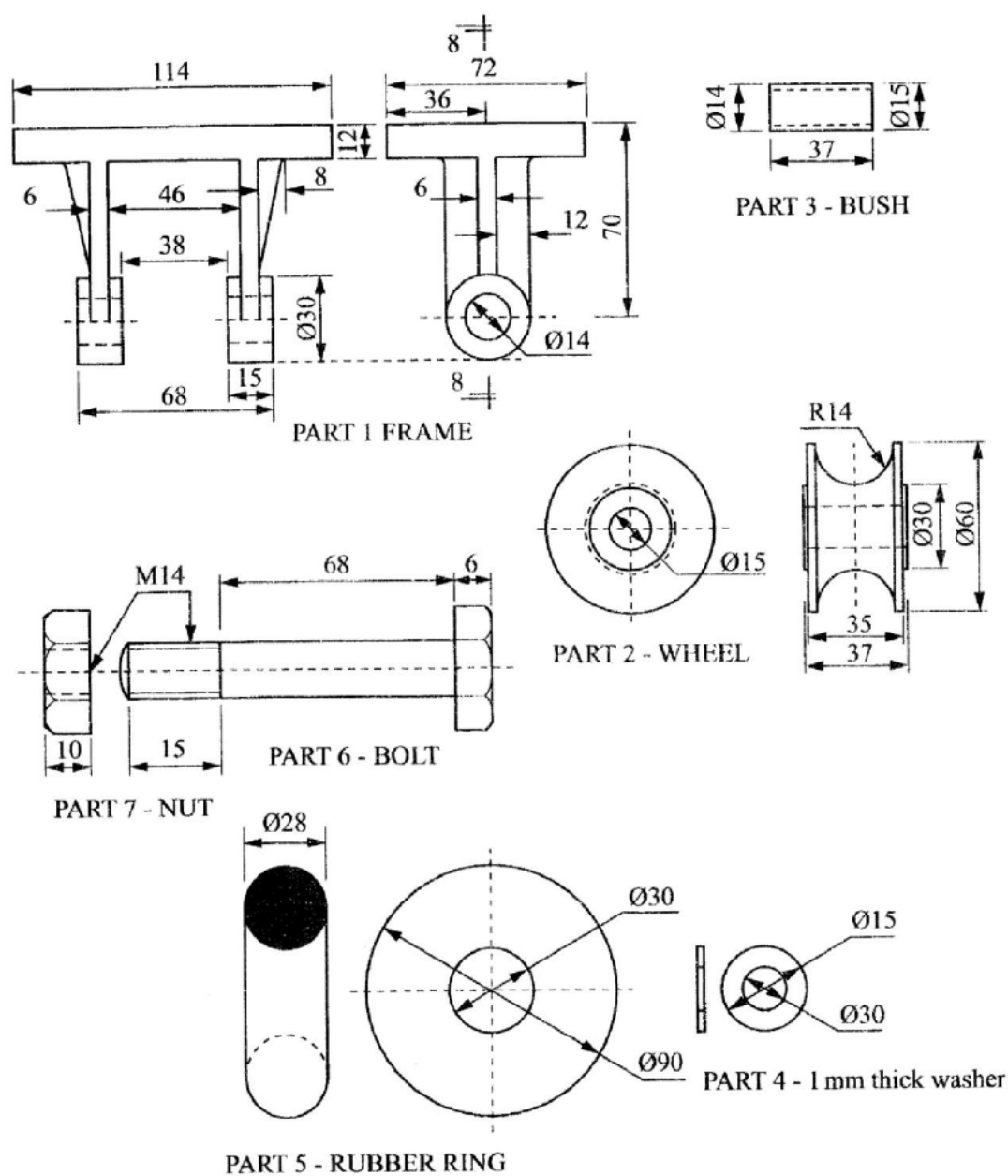
It should be answered on the A3 paper provided.

- 11.** **Figure 7** shows parts of a machine component drawn in first angle projection. Assemble the parts and draw, FULL SIZE, the following:

- (a) sectional front elevation along the cutting plane B-B;
- (b) end elevation.

Insert three leading dimensions.

Unspecified dimensions are left to the candidate's discretion. Hidden details are not required.



-- Dimensions in mm

-- Drawing not to scale

Figure 7



916104

Kenya Certificate of Secondary Education, 2016
449/1

Turn over



SECTION C (30 marks)

*Answer any **two** questions from this section on the A3 paper provided.*

- 12.** In the mechanism shown in **Figure 8**, the crank EF rotates about centre E while GH oscillates about G .

Plot the locus of point P for one complete revolution of EF .

(15 marks)

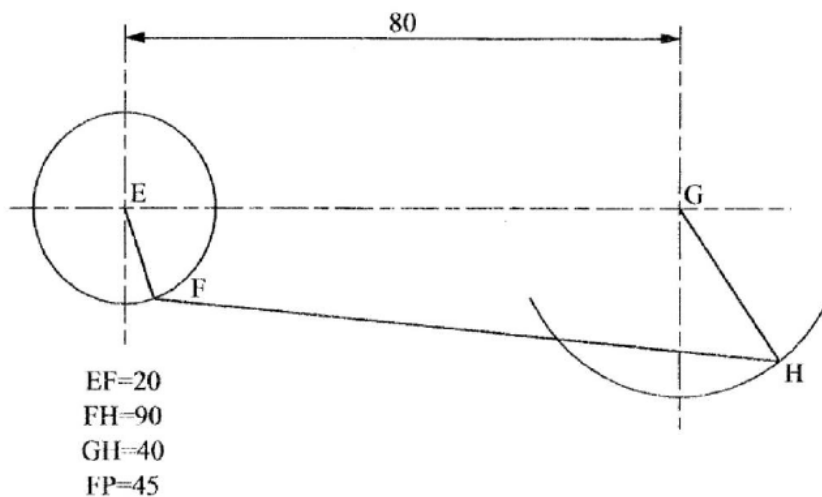


Figure 8



13. **Figure 9** shows two intersecting square tubes A and B drawn in first angle projection.

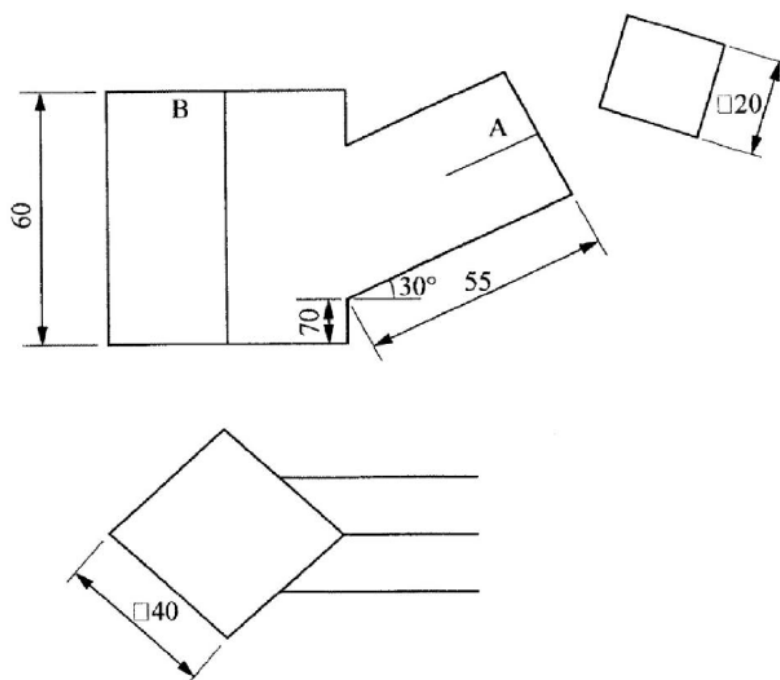


Figure 9

- (a) copy the figure and complete:

- (i) the front elevation;
- (ii) the plan.

- (b) Draw the development of tube B.

(15 marks)



916104

Kenya Certificate of Secondary Education, 2016
449/1

Turn over



14. **Figure 10** shows an inclined plan of a block and its front elevation.

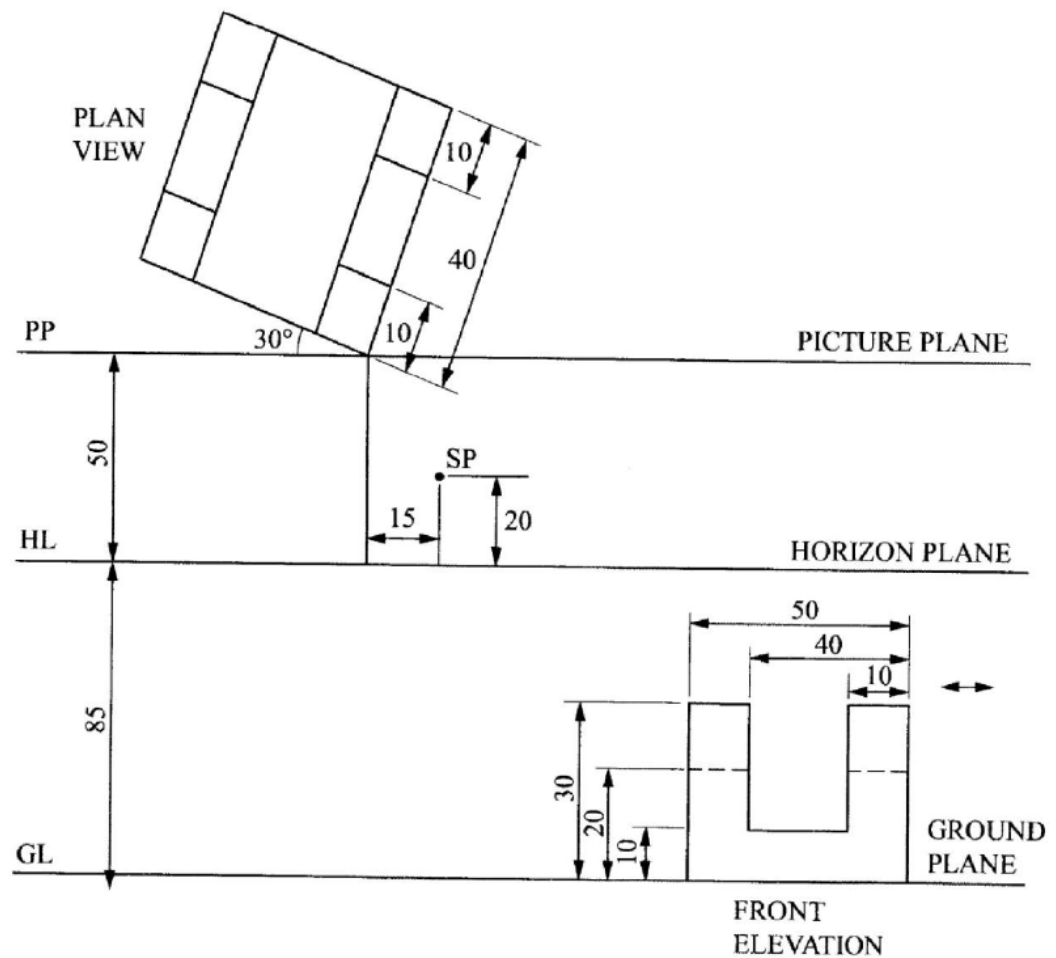


Figure 10

Copy the given layout and draw the two point perspective of the block showing all construction details. (15 mark)